

Mathematical Foundations of Political Methodology

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- Quadratic Form
- Positive Definiteness
- Multivariate Optimization

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Quadratic Form

Before we proceed our discussion to multivariate optimization, we introduce some concepts related to it. The first important concept is positive/negative definiteness.

Quadratic Form

- A *quadratic form* is a homogenous quadratic polynomial in n variables
 - A *homogeneous polynomial* is a polynomial whose nonzero terms all have the same degree: in words, the sum of the exponents of each term is constant
 - A *homogenous quadratic polynomial* is a polynomial whose terms all have degree 2, see following examples:
 - ax^2
 - $ax^2 + bxy + cy^2$
 - $ax^2 + by^2 + cz^2 + dxy + exz + fyz$ (with coefficients a, b, c, d, e, f)
 - $ax_2 + by$ is not a homogenous quadratic polynomial because one of the terms doesn't have degree 2

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Quadratic Form

Consider the two variable quadratic form

$$q(x, y) = ax^2 + bxy + cy^2$$

We can re-write it as:

$$q(x, y) = \mathbf{z}^T \mathbf{A} \mathbf{z}$$

where

$$\mathbf{z} = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } \mathbf{A} = \begin{bmatrix} a & s \\ t & c \end{bmatrix}$$

with $s + t = b$

If we impose the condition that $\mathbf{A}$ is symmetric (so that $\mathbf{A} = \mathbf{A}^T$, or $a_{ij} = a_{ji}$ for all i, j) then we have $s = t = \frac{b}{2}$

There is therefore a unique symmetric matrix $\mathbf{A}$ satisfying $q(x, y) = \mathbf{z}^T \mathbf{A} \mathbf{z}$

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There is therefore a unique symmetric matrix $\mathbf{A}$ satisfying $q(x, y) = \mathbf{z}^T \mathbf{A} \mathbf{z}$

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This generalizes:

Let $q(\mathbf{x})$ be a quadratic form in n variables, $x_1, \dots, x_n$

Then $q(\mathbf{x})$ is equivalent to the expression

$$\mathbf{x}^T \mathbf{A} \mathbf{x}$$

for a unique symmetric $n \times n$ matrix $\mathbf{A}$ and $\mathbf{x}^T = [x_1, x_2, \dots, x_n]$

Takeaway point: A symmetric $\mathbf{A}$ is uniquely determined by the coefficients in $q(\mathbf{x})$

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Positive Definiteness

For a given quadratic form $q(\mathbf{x})$ we can ask whether $q(\mathbf{x}) > 0$ (or less stringently, $q(\mathbf{x}) \geq 0$) for every non-zero $\mathbf{x}$

If $q(\mathbf{x}) > 0$ for every non-zero $\mathbf{x}$ then $q(\mathbf{x})$ is *positive definite*

- Example $q(\mathbf{x}) = x_1^2 + x_2^2 + \dots + x_n^2$

If $q(\mathbf{x}) \geq 0$ for every $\mathbf{x}$ then $q(\mathbf{x})$ is *positive semidefinite*

- Example $q(\mathbf{x}) = x_1^2 + x_2^2 + x_3^2 - 2x_2x_3$

- Notice that we can rewrite it as: $q(\mathbf{x}) = x_1^2 + (x_2 - x_3)^2 \geq 0$ but $q(0, 1, 1) = 0$

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If $q(\mathbf{x}) < 0$ for every non-zero $\mathbf{x}$ then $q(\mathbf{x})$ is *negative definite*

If $q(\mathbf{x}) \leq 0$ for every $\mathbf{x}$ then $q(\mathbf{x})$ is *negative semidefinite*

Most quadratic forms are neither positive nor negative semidefinite

Example: $q(x, y) = xy$

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Positive Definiteness

We can test if a quadratic form is positive (negative) (semi)definite by testing the associated symmetric matrix $\mathbf{A}$ that satisfies $q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$

If $\mathbf{A}$ is 2×2 , and recall, $\mathbf{A} = \begin{bmatrix} a & c \\ c & d \end{bmatrix}$, then:

- $\mathbf{A}$ is PD iff $a, d > 0$ and $\det \mathbf{A} > 0$
- $\mathbf{A}$ is PSD iff $a, d \geq 0$ and $\det \mathbf{A} \geq 0$
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Multivariate Optimization

Now, we can proceed our discussion to optimize multivariate functions

- Parameters $\beta = (\beta_1, \beta_2, \dots, \beta_n)$ such that $f(\beta|\mathbf{X}, \mathbf{Y})$ is maximized
- Policy $\mathbf{x} \in \mathbb{R}^n$ that maximizes $U(\mathbf{x})$
- Weights $\pi = (\pi_1, \pi_2, \dots, \pi_K)$ such that a weighted average of forecasts $\mathbf{f} = (f_1, f_2, \dots, f_k)$ have minimum loss

$$\min_{\pi} = -\left(\sum_{j=1}^K \pi_j f_j - y\right)^2$$

We'll describe analytic and computational approaches to optimization

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Multivariate Optimization

Definition

Let $\mathbf{x} \in \mathbb{R}^n$ and let $\delta > 0$. Define a *neighborhood* of $\mathbf{x}$, $B(\mathbf{x}, \delta)$, as the set of points such that,

$$B(\mathbf{x}, \delta) = \{\mathbf{y} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{y}\| < \delta\}$$

Multivariate Optimization

Definition

Suppose $f : X \rightarrow \mathbb{R}$ with $X \subset \mathbb{R}^n$. A vector $\mathbf{x}^* \in X$ is a *global maximum* if, for all other $\mathbf{x} \in X$

$$f(\mathbf{x}^*) \geq f(\mathbf{x})$$

A vector $\mathbf{x}^{local}$ is a *local maximum* if there is a neighborhood around $\mathbf{x}^{local}$, $Q \subset X$ such that, for all $\mathbf{x} \in Q$,

$$f(\mathbf{x}^{local}) \geq f(\mathbf{x})$$

Gradient

Definition

Suppose $f : X \rightarrow \mathbb{R}^n$ with $X \subset \mathbb{R}^1$ is a differentiable function. Define the gradient vector of f at $\mathbf{x}_0$, $\nabla f(\mathbf{x}_0)$ as,

$$\nabla f(\mathbf{x}_0) = \left(\frac{\partial f(\mathbf{x}_0)}{\partial x_1}, \frac{\partial f(\mathbf{x}_0)}{\partial x_2}, \frac{\partial f(\mathbf{x}_0)}{\partial x_3}, \dots, \frac{\partial f(\mathbf{x}_0)}{\partial x_n} \right)$$

Gradient First Order Condition

Theorem

Suppose $f : X \rightarrow \mathbb{R}^1$, $X \subset \mathbb{R}^n$. Suppose $\mathbf{a} \in X$ is a local extremum. Then,

$$\begin{aligned}\nabla f(\mathbf{a}) &= \mathbf{0} \\ &= (0, 0, \dots, 0)\end{aligned}$$

- Proof (intuition): same as one dimensional case (left-hand, right hand), just do it dimension by dimension
- Critical Values:
 - 1) Maximum
 - 2) Minimum
 - 3) Saddle point
- Second Derivative Test!

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Second Order Conditions: Hessian

Definition

Suppose $f : X \rightarrow \mathbb{R}^1$, $X \subset \mathbb{R}^n$, with f a twice differentiable function. We will define the Hessian matrix as the matrix of second derivatives at $\mathbf{x}^* \in X$,

$$\mathbf{H}(f)(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}^*) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}^*) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}^*) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

General test $\rightsquigarrow$ Two Dimensional Test $\rightsquigarrow$ Example

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General test $\rightsquigarrow$ Two Dimensional Test $\rightsquigarrow$ Example

Second Derivative Test

Many ways to assess definiteness $\rightsquigarrow$ use determinant

Theorem

Two Dimensional, Second Derivative Test. Suppose $f : X \rightarrow \mathbb{R}$ with $X \subset \mathbb{R}^2$ and f twice differentiable. Write the Hessian of f at a critical value $\mathbf{a}$,

$$\mathbf{H}(f)(\mathbf{a}) = \begin{pmatrix} A & B \\ B & C \end{pmatrix}$$

Then, we can conduct the second derivative test as:

- $AC - B^2 > 0$ and $A > 0 \rightsquigarrow$ positive definite $\rightsquigarrow \mathbf{a}$ is a local minimum
- $AC - B^2 > 0$ and $A < 0 \rightsquigarrow$ negative definite $\rightsquigarrow \mathbf{a}$ is a local maximum
- $AC - B^2 < 0 \rightsquigarrow$ indefinite $\rightsquigarrow$ saddle point
- $AC - B^2 = 0$ inconclusive

Second Derivative Test

Many ways to assess definiteness $\rightsquigarrow$ use determinant

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Multivariate Recipe

- 1) Calculate gradient
- 2) Set equal to zero, solve system of equations
- 3) Calculate Hessian
- 4) Assess Hessian at critical values
- 5) Boundary values? (if relevant)

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Example 1: A Simple Optimization Problem

Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ with

$$f(x_1, x_2) = 3(x_1 + 2)^2 + 4(x_2 + 4)^2$$

Calculate gradient

$$\nabla f(\mathbf{x}) = (6x_1 + 12, 8x_2 + 32)$$

$$\mathbf{0} = (6x_1^* + 12, 8x_2^* + 32)$$

We now solve the system of equations to yield $x_1^* = -2$ and $x_2^* = -4$

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$$\mathbf{H}(f)(\mathbf{x}^*) = \begin{pmatrix} 6 & 0 \\ 0 & 8 \end{pmatrix}$$

$\det(\mathbf{H}(f)(\mathbf{x}^*)) = 48$ and $6 > 0$ so $\mathbf{H}(f)(\mathbf{x}^*)$ is positive definite. local minimum

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Example 2: Two Dimensional Ideal Points

Suppose legislators are considering legislation $\mathbf{x} \in R^2$. And suppose legislator i has utility function $U_i : R^2 \rightarrow \mathbb{R}$,

$$U(\mathbf{x})_i = -(x_1 - \mu_1)^2 - (x_2 - \mu_2)^2$$

What is legislator i 's optimal policy?

$$\nabla f(\mathbf{x}) = (-2(x_1 - \mu_1), -2(x_2 - \mu_2))$$

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$$-2(x_1^* - \mu_1) = 0$$

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So, $-2 * -2 - 0 = 4 > 0$ and $-2 < 0 \rightsquigarrow$ **negative definite, maximum**
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